

Mid-air Haptic Texture Exploration in VR

Orestis Georgiou*, Jonatan Martinez, Abdenaceur Abdouni, and Adam Harwood

Ultraleap Ltd, Bristol, UK

ABSTRACT

Mid-air haptic feedback technology has traditionally been used to enhance gesture input interactions such as pressing a holographic button, swiping through a menu, or when grasping a virtual object. Here we present a VR research demo that expands such interactions to include for active haptic exploration. In the demo, the user can explore a virtual object using both hands and feel its intrinsic properties such as consistency and texture. The paper describes the physical apparatus used and its reasoning, the haptic rendering techniques leveraged, the research questions we plan to explore, and its relevance to applications such as VR shopping.

Keywords: Mid-air haptics, VR, haptic texture rendering.

Index Terms: H.5.1 [Human computer interaction (HCI)]: Interaction devices—Haptic devices; H.5.2 [Human computer interaction (HCI)]: Interaction paradigms—Virtual reality

1 INTRODUCTION

Advances in machine vision have enabled accurate, robust, and fast hand tracking algorithms to run on consumer devices empowering novel HCI applications, including training, gaming, tele-operation, natural user interfaces, and e-commerce. For example, typical freehand gesture interactions in VR immersive shopping environments include selection, rotation, reveal menu, enlarge/shrink object, back/next item, try-on, and add to shopping cart [1]. Note that while these gestures are often referred to as *natural* they are actually quite artificial as they do not exist in the real world. Instead, these gestures are semantic analogues and representations of the intended inputs, actions or functions.

Mid-air haptics uses phased arrays of ultrasound speakers to project high acoustic pressure fields onto the users' hands with a high spatial and temporal resolution [2]. To date, this technology has mostly been applied to deliver haptic feedback during virtual button press or selection gestures and to enhance the user experience [3]. Increasingly however, mid-air haptics have been leveraged to enable the use of more natural gesture interactions that also exist in the real world and capitalize on the motor capabilities of the hand for dexterous manipulation and exploration of virtual objects, and the extraction of their intrinsic properties such as object size, shape, compliance, and texture [4][5][6]. Recall that in real life, haptic exploration and identification of physical objects tends to follow strategic patterns or routines, e.g., we apply pressure to infer hardness, static contact for temperature, contour following for shape, and lateral motion for texture [7]. Motivated by this, we argue that mid-air haptics can enable users to apply natural haptic exploration routines to interactions in VR/XR environments.

In this paper, we present a novel multisensory VR research demo that allows for freehand haptic *texture* exploration of virtual objects. The demo enables us to research new 3D interaction styles for haptic exploration, compare haptic rendering algorithms, evaluate user experience (UX) and immersion, investigate visuo-

haptic and audio-haptic congruence, and create application-specific environments, e.g., VR shopping, gaming, training, heritage, etc.

Previous experimental results in VR shopping applications rated freehand-gesture-based interactions higher in terms of task load, UX, and presence without the loss of performance (i.e., speed and error count) compared to traditional immersive VR interaction techniques, such as the virtual handle controller and ray-casting techniques [1]. Our demo aims to further enhance such capabilities through the haptic texture exploration of virtual objects for a variety of immersive applications and use cases, not just VR shopping.

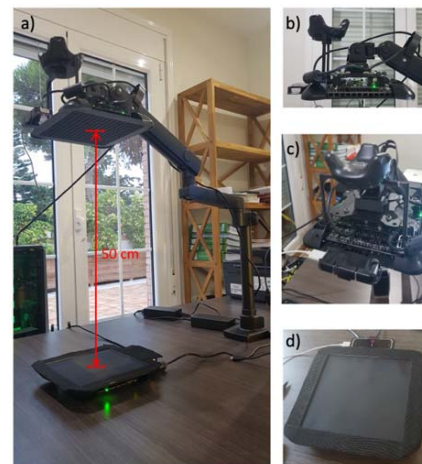


Figure 1: a) Demo setup consisting of two USX in sandwich arrangement 50cm apart. b) and c) Close up pictures of the top USX and 3D printed mount holding a VIVE tracker. d) Bottom USX covered with black felt material to reduce infrared reflections.

2 DEMONSTRATION SETUP

Figure 1 shows the demonstration setup. To maximize object haptic exploration, we used two Ultraleap Stratos eXplore devices (USX) arranged in a sandwich configuration. The bottom array is placed on a table while the top array is facing downwards, mounted on a VESA support, and rotated 180° in Z axis with the Leap Motion Controller (LMC) mounted closest to the user. The separation between the two USXs is 50 cm but could be adjusted within the tracking range of the LMC (25-60 cm). To limit infrared light contamination and improve LMC performance, we cover the bottom array with black felt (an acoustically transparent material). The top array is used to define where everything is located within the VR scene using a VIVE tracker held using a custom 3D printed support that slots into the USX. The bottom USX is placed directly below the top USX using a ruler. The distances and orientations between the 2 USX, the VIVE tracker and the LMC are all pre-calibrated within the Unity3D scene which runs on a host PC.

3 VR SCENE AND FREEHAND EXPLORATION

In the VR scene, the user sees a virtual object floating between two projectors. The projectors represent the physical locations of the two USXs and should not be touched since that could ruin the scene calibration. The user also sees their two avatar hands as is usual in LMC implementations. There is no physics-based hand-

* orestis.georgiou@ultraleap.com

object interaction implemented, i.e., the avatar hands will pass through the virtual object with no pseudo-haptic visual effects. Instead, different virtual objects will react differently depending on the user's gestures.

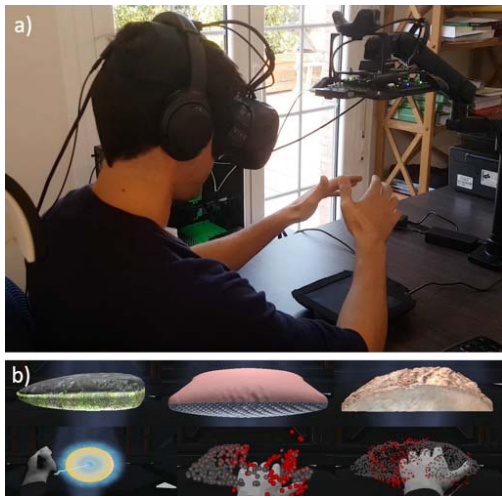


Figure 2: a) Freehand haptic exploration of the objects shown in b).

Figure 2 shows how a user would interact with the different objects and how they react to his touch: a rock covered with moss on one side; a pillow made of two different fabrics; a sliced tree trunk exposing its outside bark and its smoothly sanded inside; a magical lightning orb; a point-cloud shaped like a pillow made up of a low (high) number of big (small) particles. The floating rock, the pillow, and the tree trunk slightly displace in the direction that the avatar hand pushes. Also, a rubbing sound is played in a loop while being modulated in volume and pitch according to the tangential avatar hand speed. When the avatar hands are in proximity to the magic orb, electric sparks shoot out making high voltage sounds. The particle clouds are displaced and turn red when pushed away by the avatar hand and then return to their original location and color through attractive forces. Finally, each of these objects has a different mid-air haptic rendering applied thus inducing a different texture feeling. The object displacement, the rubbing sounds, and the applied mid-air haptic effects can be toggled on/off via hotkeys for research and comparison purposes.

4 HAPTIC RENDERING

We have adapted the haptic rendering algorithms of [6] to the 3D objects shown in the first row of Figure 2 b). Namely, we apply the texture scanning pipeline of [6] to the unwrapped image of a (moldy) rock, the different fabrics, and wood types, to predict their perceived visual roughness ratings which are then mapped to a haptic parameter value. The latter corresponds to the draw frequency of an STM circle sensation that is projected onto the user's palm or fingertips during haptic exploration of the object. The STM circle path used has radius 3 cm, and the draw frequency ranges between 20-80 Hz for a rough or smooth texture sensation, respectively [8]. The haptic sensation is generated by the USX device that has the best line of sight (LOS) of the user's palm, inferred via the palm's normal vector. Also, the intensity of the haptic texture sensation, i.e., the acoustic output power of the USX, is modulated by the speed of the hand. A static hand (rubbing hand motion) intersecting the virtual object will have a low (high) intensity. Finally, the hand-object intersection coordinates, the palm's normal vector, and the hand speed are obtained from the LMC API at a 60 Hz refresh rate. Note that the demo setup could

be expanded to utilize 3+ USXs, thus improving the interaction region, the haptics quality, and the LOS to the user's palms.

The haptic rendering induced when interacting with the orb uses two haptic sensations depending on the hand to orb-center distance: *RandomPalmPoints* (0-5 cm) and *RandomTravelPoint* (5-10 cm). In *RandomPalmPoints*, four AM haptic points modulated at 200 Hz are imprinted on the palm at random locations that change every 0.2 seconds causing a fluttering sensation. In *RandomTravelPoint*, a single AM haptic points modulated at 200 Hz traverses from palm center to fingertip along a randomly chosen finger every 1 second.

The haptic rendering induced when interacting with the point-cloud uses the *RandomPalmPoints* sensation with a randomization period of 0.2 or 0.1 seconds for big or small particles, respectively.

The above mid-air haptic rendering techniques are not novel. What is novel is their dynamic parameter couplings to haptic exploration gestures such as hand speed, akin to the texture rendering system proposed by [9]. This is also the first two-hand VR demo that uses multiple USX devices to simultaneously display different textured information, e.g., when touching the two sides of the rock, pillow, or tree-trunk. Finally, the resulting multisensory VR textures demo provides a compelling and novel simulation of haptic exploration of real, magical, and point-cloud objects. It also forms the basis for future human subject research studies aimed at improving haptic rendering techniques and understanding visuo-haptic and audio-haptic congruence effects.

5 CONCLUSION

Haptic exploration involves exploratory procedures, active touch patterns that are specific to the task in that they optimize the extraction of information the perceiver needs to obtain [7]. In VR, this has to date been mediated through the use of hand-held haptic controllers or gloves. In this demo paper we have leveraged ultrasound mid-air haptics to enable the freehand haptic exploration of virtual objects and feel their consistency and texture [6]. The described research demo is designed to facilitate future human perceptual studies and the development of immersive applications.

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